



Introduction to R and Reproducible Research

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Jan 13, 2025

Today

- Syllabus
- Why learn R?
- Essential programs/packages
 - Free resources
 - R as a calculator

Course Info

Credits: Two

Meets: MW, 10:35-11:30am

Meeting Location: 309 Baker

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Office hours: W 2-3pm

Teaching Asst.: NA

Email:

3/23

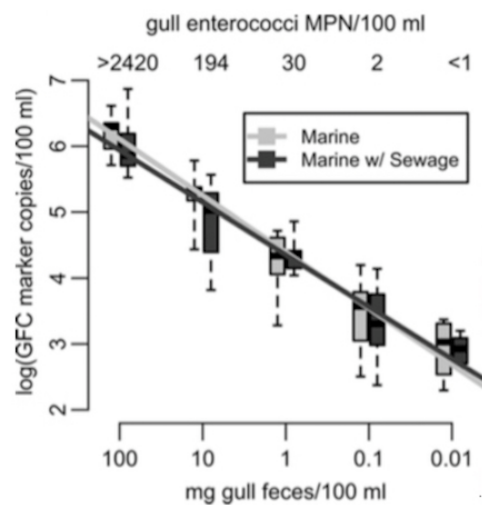
Go to Syllabus

4/23

My Research and R

5/23

My Research and R



- Not an expert in R or statistics, but ...
- using R most days for 15 years
- plotting, stats, data wrangling
- via RStudio
- Have not published a paper or written a grant *without* using R!

6/23

This is hard! Why do this?

- Save your time
- building off of previously designed methods (by you or others)
- Improves the appearance of your work (R + other tools)
- Actually helps you perform *better* research
- take 'bad' shortcuts less often
- permits thorough criticism
- Organizes your work (which helps you do all of the above)
- More research opportunities
- permits the analysis of 'large' data (and 'big' data)
- Big picture: Limits reproducibility/replicability `crisis'
- Future requirement of scientific journals

7/23

Some Important Learning Outcomes

- With R, you often plan and build a foundation to your analyses
- Not only *learn R* but also *practice R*
- Choose to use R even though it might cost you a little time in the beginning
- Encourage others to learn R

8/23

R and Associated Programs

- R
- R Studio
- knitr
- RMarkdown

9/23

What is R?

- Programming *language*
 - it has its own syntax
 - if you learn it, you can make R do stuff for you
- Programming *environment*
 - Not just a list of tools
 - but a “fully planned and coherent system”
 - permits creation of new tools and new clusters of tools called *packages*

10/23

What does R do?

- statistics
- plotting
- data handling, manipulation, and efficient storage
- you can make R do pretty much anything (see [r.bloggers](#))

11/23

Advantages of R

- Free, open source
- Seemingly endless number of packages for pretty much any type of data analysis you can imagine
- the *latest* analysis tools
- Large, usually helpful, community of users (see [stackoverflow](#))
- Tools for communicating results (easily accessible through RStudio)

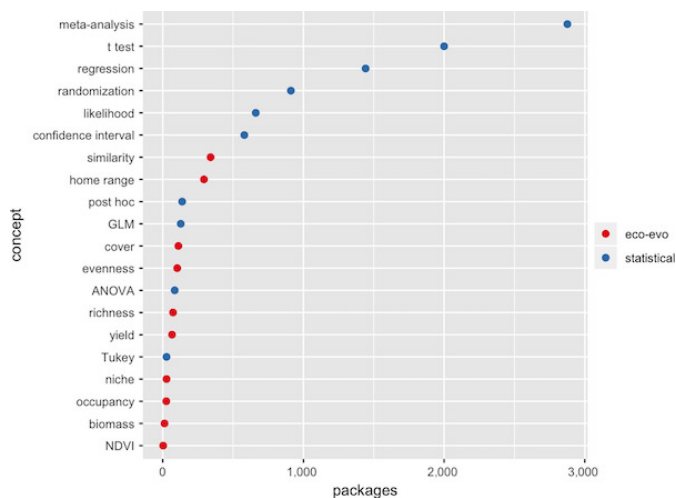
12/23

Disadvantages of R

- Much of the written code you'll see is sloppy
 - Authors are focused on the results. Hasty.
 - Diverse backgrounds
- Inconsistent
 - Many packages for the same purpose
 - Many special cases to remember
- Not that fast. Poorly written code can be really slow.

13/23

Redundant R Packages



from Lortie et al. 2020

14/23

Accessing R

- R
- R Studio
- Terminal (Mac)

15/23

R as a Calculator

- arithmetic
- basic functions `mean()`, `sd()`, `Sys.time()`
- help with specific functions
- `demo()` to check out R capabilities

16/23

Accessing More R Resources

- CRAN
- Task Views
- Quick-R
- R-bloggers
- Advanced R
- [stackoverflow](#)
- Google (helps to know specific terms)

17/23

R Studio

- Integrated Development Environment
- Offers quick point-and-click access to stuff you otherwise have to type in longhand
 - files
 - plots
 - packages
 - creating documents
 - etc
- Requires R still

18/23

Go to R Studio

19/23

RMarkdown

- Drafting language designed to be simple and intuitive
- Allows embedding of R code

20/23

knitr

- 'knits' R code and output (plots, etc) into Rmarkdown
- Produces many different types of output
 - pdf
 - html
 - docx
- Does not do any analysis for you, just formats the output

21/23

Why learn Rmarkdown/knitr?

22/23

Questions?